Fiscal Year:	FY 2013	**Task Last Updated:**	FY 04/24/2013
PI Name:	Ianzini, Fiorenza Ph.D.		
Project Title:	Mechanisms of Cell Survival Following Space Radiation-Induced Mitotic Catastrophe: Implications for Cancer Risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1009	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	06/01/2012	**End Date:**	01/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 1/31/2013, per NSSC information (Ed., 12/21/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mackey, Michael (University of Iowa)		
Grant/Contract No.:	NNX12AG50G		
Performance Goal No.:			
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	[Ed. note, 6/13/2012: Note this is a continuation of project with the same title, "Mechanisms of Cell Survival Following Space Radiation-Induced Mitotic Catastrophe: Implications for Cancer Risk," grant NNX11AB60G. See that project for previous reporting] The overall goal of our studies is to characterize novel molecular mechanisms involved in radiation carcinogenesis. Our studies concentrate on the role played by altered cell proliferation, aberrant mitoses, and aneuploidy/polyploidy cell formation and test the hypothesis that cells undergoing mitotic catastrophe (MC – these cells are large multinucleated, polyploid cells) after exposure to charged particle radiation are able to escape death, to undergo depolyploidization, to give rise to cell progenies that regain a degree of genomic stability sufficient for them to survive, to proliferate, and to acquire transformation competence thus becoming a focal spot for cancer initiation.		

Task Description:

Our studies will be extended during this new grant period by using regimens of fractionated irradiations. In particular, during the NSRL (NASA Space Radiation Laboratory at Brookhaven) Fall 2012 Campaign, we propose to expose normal human cells to fractionated iron ions at various energies and doses. Live cell imaging, immunofluorescence, and Western Blotting, experiments will be carried out to determine the overall effect of fractionated charge particle exposures on production of polyploid/aneuploid cells, their genetic signatures, their potential for survival and the fate of their progenies. Moreover, transformation experiments using the survivors of the polyploid cells will be designed that will help to determine the long-term detrimental effects of fractionated radiation as it pertains to cancer risk. The data obtained will be compared with those we already have on acute radiation exposures. We believe that this approach will advance our understanding of the effects of low dose and low dose-rate exposures delivered over a period of time on normal cells as it pertains to the end-points discussed above (all hallmarks of cancer phenotype and genotype) and thus, will help in identifying the contribution of these events to cancer risk from space radiation exposures.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Final Task Book Report - for the period of 06/01/2012 - 01/31/2013 - (6 months cost-extension)
Review of Progress

The last 6 months of funding under this project were spent analyzing data for publication and conducting measurements for radiation-induced changes at the genetic and epigenetic level on samples that were irradiated and collected during previous irradiation campaigns.

Summarizing our overall findings we can affirm that one of the most important discoveries of our studies is that we demonstrate that all the cell lines we have analyzed (murine or human, embryonic, fetal or adult) undergo space radiation-induced mitotic catastrophe (MC) and that, on the contrary of what it might be expected, a percentage of these cells are able to survive for weeks post-irradiation. These results, coupled with findings on cancer cells (Ianzini et al. 2009) that demonstrated that irradiated cells undergo MC, become polyploid, and are able to escape death by activating processes of depolyploidization that utilize uncharacteristic pathways of cell divisions (namely activation of meiotic genes during multipolar divisions), prompted us to hypothesize that similar processes might occur in normal cells exposed to space radiation.

The occurrence of these processes, associated with the production of persistent DNA damage (also found in our irradiated cells), is of great importance as it would imply that normal cells that receive a dose of space radiation and undergo MC can escape death by re-organizing their genomic composition. If these cells are also able to proliferate indefinitely (as it happens for irradiated cancer cells – Ianzini et al. 2009), their progeny would likely have a disarrayed DNA that could be prone to further mutations and ultimately render this cell progeny more susceptible to carcinogenesis.

We set at determining the mechanisms that favor genomic reorganization in the irradiated cells and, in the past few months, we also investigated genetic and epigenetic changes that may play a role in the carcinogenesis process. In this contest, we demonstrated:

- activation of both meiosis-specific genes and specific markers of cancer stem cells in the irradiated cells (Kosmacek et al., in preparation); these findings point towards the novel notion that unusual pathways of cell division and proliferation are involved in genomic reorganization in irradiated cells that escape death;

- disruption of normal arrangement of loci in chromosomes that have signatures for different types of cancers (karyotyping experiments in collaboration with Dr. Daila Gridley from Loma Linda University); these aberrations are found in both the original irradiated cells and their progeny highlighting the fact that they propagate to subsequent generations and may hinder the overall genetic stability of the cellular system that could then become more prone to mutation and transformation;

- dramatic decrease in Lsh mRNA and protein levels associated with persistent gamma-H2AX foci (the lymphoid-specific helicase (Lsh) is a gene that is preferentially associated with pericentric heterochromatin and whose suppression leads to CpG demethylation and disruption of proper DNA conformation; these are important functions in our system because changes in cell ploidy, aberrant chromatin organization, aberrant mitoses, and skewed cell proliferation are associated with disruption of normal organization of pericentric heterochromatine (Bourgo et al. 2009)).

Taken together, our findings outline the innovative concept that normal human cells exposed to charged particles mimicking space radiation become polyploid - a tumor hallmark -, are minimally affected by cell death, and a percentage of them undergo successful depolyploidization and form progenies. The polyploidization/depolyploidization (reduction division) events are accompanied by morphologic, phenotypic, genetic, and epigenetic changes in which the cells' DNA content greatly varies, the cells' chromatin condensation status is greatly altered, proteins involved in DNA repair/remodeling and meiosis-specific proteins involved in pairing of homologous chromosomes are found highly expressed, and proteins involved in the proper establishment and maintenance of DNA methylation at pericentric heterochromatin are greatly depressed. The latter has been shown to lead to disruption of normal organization of chromatin, which in turn is associated with changes in cell ploidy, aberrant chromatin organization, aberrant mitoses, and skewed cell proliferation; all events that occur in our irradiated cells.

Our results provide a mechanistic link between radiation-induced changes in DNA methylation and altered cell proliferation, aberrant mitoses, chromatin condensation statuses, and changes in ploidy in normal human irradiated cells, and support the hypothesis that the totality of these modifications are events linked to genetic and epigenetic modification that might contribute to the overall cellular genetic instability (without greatly affecting cell death) of the exposed populations and their progeny and, thus, in turn contribute to augmented cancer risk from space radiation.

References

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Ianzini F., Kosmacek E.A., Nelson E.S., Napoli E., Erenpreisa J., Kalejs M., Mackey M.A. 2009. Activation of

	meiosis-specific genes is associated with depolyploidization of human tumor cells following radiation-induced mitotic catastrophe. Cancer Res., 69, 2296-2304. Kosmacek E.A, Napoli E., Mackey M.A., Ianzini F. 2013. Iron ion radiation induces changes in ploidy, persistent activation of DNA repair enzymes and expression of the meiotic-specific protein SYCP3 in normal human fibroblasts. In preparation.
Bibliography Type:	Description: (Last Updated: 02/12/2021)
Articles in Peer-reviewed Journals	Meng X, Laidler LL, Kosmacek EA, Yang S, Xiong Z, Zhu D, Wang X, Dai D, Zhang Y, Wang X, Brachova P, Albitar L, Liu D, Ianzini F, Mackey MA, Leslie KK. "Induction of mitotic cell death by overriding G2/M checkpoint in endometrial cancer cells with non-functional p53." Gynecologic Oncology. 2013 Mar;128(3):461-9. Epub 2012 Nov 9. http://dx.doi.org/10.1016/j.ygyno.2012.11.004 ; PubMed PMID: 23146687 , Mar-2013